

- 1 A fair 6-sided dice with faces labelled 1, 2, 3, 4, 5, 6 is thrown repeatedly. The random variable X denotes the number of throws required to obtain a 6.

(a) Find $P(X=7)$.

1/1

$$X \sim \text{Geo}\left(\frac{1}{6}\right)$$

$$\left(\frac{5}{6}\right)^6 \times \frac{1}{6} = 0.0558$$

Ans 0.0558

Correct method and accurate calculation for the geometric probability $P(X=7)$.

0/2

(b) Find $P(3 \leq X \leq 5)$.

$$P(X \leq 5) = 1 - \left(\frac{5}{6}\right)^5$$

$$= 0.598$$

$$P(X \leq 2) = 1 - \left(\frac{5}{6}\right)^2$$

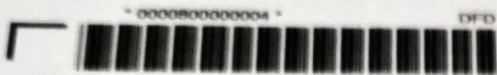
$$= 0.306$$

$$0.598 - 0.306 = 0.292$$

The interpretation of $P(3 < X < 5)$ as $P(X \leq 5) - P(X \leq 3)$ is incorrect for discrete X , it should be $P(X=4)$. Furthermore, the calculation for $P(X \leq 3)$ is incorrect (appears to be $P(X \leq 2)$).

Ans 0.292





4

- 2 Kai has a spinner with four sides, labelled 1, 2, 3, 4. When the spinner is spun, the score is the number on the side on which the spinner lands. The random variable X denotes this score. The probability distribution table for X is given below.

x	1	2	3	4
$P(X=x)$	p	0.4	$2p$	p

- (a) Find the numerical value of $\text{Var}(X)$.

[4]

$$4p + 0.4 = 1$$

$$4p = 0.6$$

$$p = 0.15$$

$$E(X) = (1 \times 0.15) + (2 \times 0.4) + (3 \times 0.3) + (4 \times 0.15)$$

$$= 2.45$$

$$E(X)^2 = (1^2 \times 0.15) + (2^2 \times 0.4) + (3^2 \times 0.3) + (4^2 \times 0.15)$$

$$= 6.85$$

$$6.85 - (2.45)^2$$

$$= 0.8475$$

$$\text{Var}(X) = 0.8475$$

All steps for calculating p , $E(X)$, $E(X^2)$ and $\text{Var}(X)$ are correct and accurate.

4/4



Kai spins his spinner 10 times.

- (b) Find the probability that a score of 2 is obtained fewer than 8 times.

[3]

~~$X \sim B(10, 0.4)$~~

$$P(X \sim B(10, 0.4))$$

$$P(X < 8)$$

$$1 - (P(8) + P(9) + P(10))$$

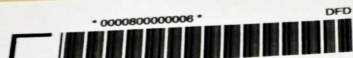
$$= 1 - ({}^{10}C_8 \times (0.4)^8 \times (0.6)^2 + {}^{10}C_9 \times (0.4)^9 \times 0.6 + {}^{10}C_{10} \times (0.4)^{10} \times 0.6^0)$$

$$= 1 - 0.0123$$

$$= 0.9877$$

3/3

Correct identification of Binomial distribution parameters, correct method using $P(X < 8) = 1 - P(X \geq 8)$, and accurate calculation.



3 Priti has two bags of discs, X and Y .

Bag X contains 8 red discs and 7 blue discs.

Bag Y contains 6 red discs and 9 blue discs.

Priti tosses a fair coin.

If she obtains a head, she chooses at random and **without** replacement two discs from bag X .

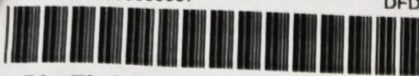
If she obtains a tail, she chooses at random and **with** replacement two discs from bag Y .

(a) Draw a tree diagram to represent this information, showing all the probabilities.

[2]

0/2

The student's answer (tree diagram) for this question is not provided in the submitted images.



(b) Find the probability that the two discs that Priti chooses are blue.

[2]

2/2

$$P(B, B) = \frac{1}{2} \times \frac{7}{15} \times \frac{6}{14} = \frac{42}{420} = \frac{1}{10}$$

$$P(B, B) = \frac{1}{2} \times \frac{9}{15} \times \frac{8}{15} = \frac{81}{450} = \frac{9}{50}$$

Correctly calculated probabilities for both paths (from Bag X and Bag Y) and summed them to find the total probability.

2/4

(c) Find the probability that the two discs that Priti chooses come from bag X given that at least one of the discs is red.

[4]

Bag X:

$$P(B, B) = \frac{1}{2} \times \frac{7}{15} \times \frac{6}{14} = \frac{42}{420}$$

$$P(R, R) = \frac{1}{2} \times \frac{8}{15} \times \frac{7}{14} = \frac{56}{420}$$

$$P(B, R) = \frac{1}{2} \times \frac{8}{15} \times \frac{7}{14} = \frac{56}{210}$$

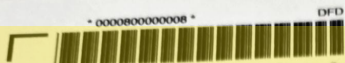
Bag Y:

$$P(R, R) = \frac{1}{2} \times \frac{6}{15} \times \frac{5}{14} = \frac{30}{420}$$

$$P(R, B) = \frac{1}{2} \times \frac{6}{15} \times \frac{9}{15} = \frac{54}{450}$$

$$P(\text{at least one red}) = \frac{8}{25} + \frac{2}{5} = \frac{8+10}{25} = \frac{18}{25}$$

$$\frac{2}{5} \times \frac{25}{18} = \frac{10}{18} = \frac{5}{9}$$



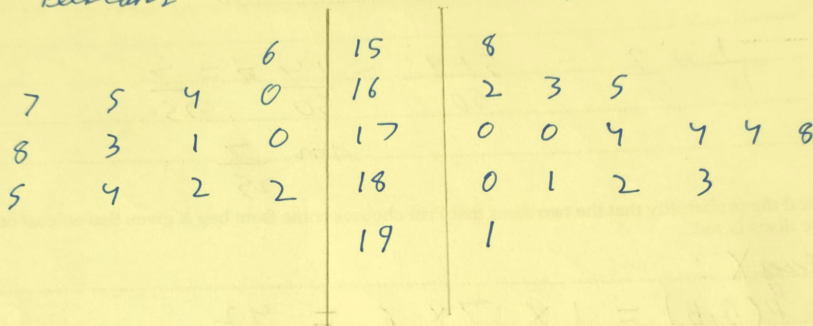
4 The heights, in cm, of 15 players from each of two sports teams, Pelicans and Swans, are given in the table.

Pelicans	156	160	164	165	167	170	171	173	176	182	182	184	185	186	187
Swans	170	180	183	165	174	158	170	181	162	178	174	163	191	182	174

- (a) Draw a back-to-back stem-and-leaf diagram to represent the heights of the players from Pelicans and Swans, with Pelicans on the left-hand side. [4]

Pelicans

Swans

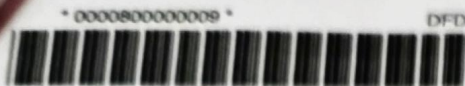


Key = $15/8/8$ 6/15/8 156 for pelicans
158 for swans

Correct stems and Swans leaves (in order). However, Pelicans leaves are not in order. The key is ambiguous and does not correctly represent the back-to-back format.



given in



(b) Find the median and the interquartile range of the heights of the Pelicans.

[3]

Median = 173

IQR = 184 - 165

= 19

0/3

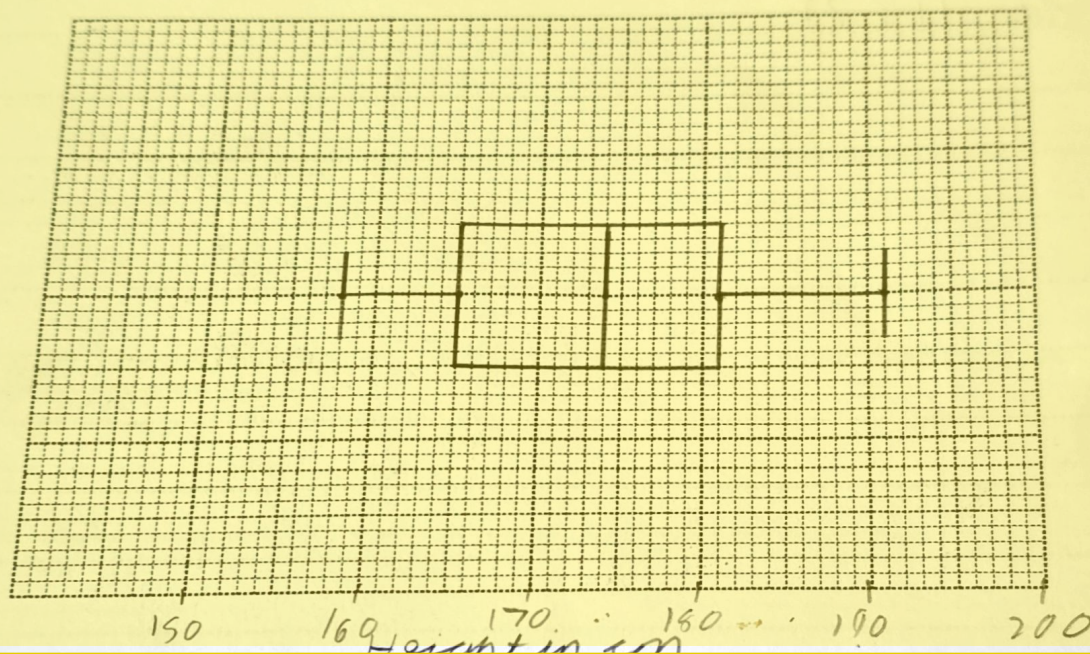
Incorrect median and quartiles. The method for IQR ($Q3 - Q1$) is correct, but applied to incorrect values.

For the Swans, the lower quartile of the heights is 165 cm, the median is 174 cm and the upper quartile is 181 cm.

2/3

(c) Represent the data shown in the back-to-back stem-and-leaf diagram by a pair of box-and-whisker plots in a single diagram.

[3]



The box plot for Swans is correctly plotted using the provided summary statistics and data min/max. The scale is appropriate. However, the Pelicans plot has errors in its median and lower quartile placement.

(d)

Make a comparison between the heights of the Pelicans players and the heights of the Swans players.

[1]

The height of Swans players are more distributed

0/1

The comparison regarding 'more distributed' (meaning wider spread) is incorrect, as the Swans have a slightly smaller IQR (16) compared to Pelicans (17).

- 5 In a group of 25 athletes, there are 8 sprinters, 5 hurdlers and 12 throwers.

(a) Find the number of different ways in which a team of 6 athletes can be selected if it consists of at least 3 sprinters, at most 2 hurdlers and at most 1 thrower. [4]

S	H	T	
3	2	1	$= {}^8C_3 \times {}^5C_2 \times {}^{12}C_1 = 6720$
4	1	1	$= {}^8C_4 \times {}^5C_1 \times {}^{12}C_1 = 4200$
5	1	0	$= {}^8C_5 \times {}^5C_1 \times {}^{12}C_0 = 280$
5	0	1	$= {}^8C_5 \times {}^5C_0 \times {}^{12}C_1 = 672$
			<u>11872</u>

Ans 11872

Correctly identified all valid combinations of sprinters, hurdlers, and throwers satisfying the conditions. Individual combination calculations ($C(n,k)$) are correct. However, there is an arithmetic error in the final sum.



A group of 8 athletes chosen from the group of 25 athletes consists of 1 sprinter, 3 hurdlers and 4 throwers. These 8 athletes stand in a row.

- (b) How many different arrangements of the 8 athletes are there if the 3 hurdlers do not stand all together? [3]

$$8! = 51 \times 6 \times 3 = 2900$$

$$\boxed{H \ H \ H} = 6! \times 3! = 4320$$

$$\text{Total arrangements} = 8! = 40320$$

$$40320 - 4320 = 36000$$

3/3

Correctly calculated total arrangements and arrangements where hurdlers are together, then subtracted to find the required probability.

- (c) How many different arrangements of the 8 athletes are there in which there are at least 2 athletes between any two hurdlers? [3]

~~H~~ — — — ~~H~~ — — —

$$6! \times 5 \times 3 = 10800$$

The method '6! x 5 x 3' is incorrect for this type of problem, and the resulting calculation is also incorrect (1080.0 instead of 10800 if 6! * 5 * 3 was intended).

- 6 A large number of runners took part in two charity runs to raise money for a new community centre.

In the first run, the times to complete the run were normally distributed with mean 46.3 seconds and standard deviation 6.4 seconds.

- (a) Find the probability that a randomly chosen runner took more than 55.1 seconds to complete the run. [3]

$$X \sim N(46.3, 6.4^2)$$

$$P(X > 55.1)$$

$$P\left(Z > \frac{55.1 - 46.3}{6.4}\right)$$

$$P(Z > 1.375)$$

$$1 - \Phi(1.375)$$

$$= 0.0845$$

3/3

Correctly identified the Normal distribution and its parameters. Correctly calculated the Z-score and used the Z-table for the final probability.

Correctly set up the probability statement for the Normal distribution. Accurately used the inverse Normal function to find the Z-value and solved for sigma.

3/3

In the second run, the times to complete the run were normally distributed with mean 39.8 seconds and standard deviation σ seconds. 10% of the runners took more than 48.6 seconds.

- (b) Find the value of σ .

$$X \sim N(39.8, \sigma^2)$$

$$P(X > 48.6) = 0.1$$

$$P\left(Z > \frac{48.6 - 39.8}{\sigma}\right) = 0.1$$

$$P\left(Z > \frac{8.8}{\sigma}\right) = 0.1$$

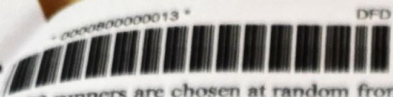
$$\frac{8.8}{\sigma} = \Phi^{-1}(0.9)$$

$$\text{Ans } 6.87$$

$$\frac{8.8}{\sigma} = 1.281$$

$$\sigma = \frac{8.8}{1.281}$$

$$\sigma = 6.87$$



150 runners are chosen at random from those who took part in the second run.

- (c) Use an approximation to find the probability that fewer than 20 of the 150 runners took more than 48.6 seconds to complete the run. [5]

$$np = 150 \times 0.1 = 15$$

$$nq = 150 \times 0.9 = 135$$

$$npq = 150 \times 0.9 \times 0.1 = 13.5$$

$$X \sim B(150, 0.1) \sim Y \sim N(15, 13.5)$$

$$P(X < 20)$$

$$P\left(Z < \frac{19.5 - 15}{\sqrt{13.5}}\right)$$

$$P(Z < 1.225)$$

$$= 0.8901$$

3/3

Correctly identified the Binomial distribution and applied the Normal approximation with continuity correction. All calculations for mean, variance, Z-score, and final probability are accurate.

Grading Summary

Total Score: 31 / 48 (64.6%)

Grade: C

Question Breakdown

1(a): 1/1 (100%) ? Correct method and accurate calculation for the geometric probability $P(X=7)$.

1(b): 0/2 (0%) ? The interpretation of $P(3 < X < 5)$ as $P(X \leq 5) - P(X \leq 3)$ is incorrect for discrete X , it should be $P(X=4)$. Furthermore, the calculation for $P(X \leq 3)$ is incorrect (appears to be $P(X \leq 2)$).

2(a): 4/4 (100%) ? All steps for calculating p , $E(X)$, $E(X^2)$ and $\text{Var}(X)$ are correct and accurate.

3(b): 3/3 (100%) ? Correct identification of Binomial distribution parameters, correct method using $P(X < 8) = 1 - P(X \geq 8)$, and accurate calculation.

4(a): 0/2 (0%) ? The student's answer (tree diagram) for this question is not provided in the submitted images.

4(b): 2/2 (100%) ? Correctly calculated probabilities for both paths (from Bag X and Bag Y) and summed them to find the total probability.

5(a): 2/4 (50%) ? Correct stems and Swans leaves (in order). However, Pelicans leaves are not in order. The key is ambiguous and does not correctly represent the back-to-back format.

5(b): 0/3 (0%) ? Incorrect median and quartiles. The method for IQR ($Q3 - Q1$) is correct, but applied to incorrect values.

5(c): 2/3 (67%) ? The box plot for Swans is correctly plotted using the provided summary statistics and data min/max. The scale is appropriate. However, the Pelicans plot has errors in its median and lower quartile placement.

5(d): 0/1 (0%) ? The comparison regarding 'more distributed' (meaning wider spread) is incorrect, as the Swans have a slightly smaller IQR (16) compared to Pelicans (17).

6(a): 3/4 (75%) ? Correctly identified all valid combinations of sprinters, hurdlers, and throwers satisfying the conditions. Individual combination calculations ($C(n,k)$) are correct. However, there is an arithmetic error in the final sum.

6(b): 3/3 (100%) ? Correctly calculated total arrangements and arrangements where hurdlers are together, then subtracted to find the required probability.

6(c): 0/3 (0%) ? The method ' $6! \times 5 \times 3$ ' is incorrect for this type of problem, and the resulting calculation is also incorrect (1080.0 instead of 10800 if $6! \times 5 \times 3$ was intended).

7(a): 3/3 (100%) ? Correctly identified the Normal distribution and its parameters. Correctly calculated the Z-score and used the Z-table for the final probability.

7(b): 3/3 (100%) ? Correctly set up the probability statement for the Normal distribution. Accurately used the inverse Normal function to find the Z-value and solved for sigma.

7(c): 3/3 (100%) ? Correctly identified the Binomial distribution and applied the Normal approximation with continuity correction. All calculations for mean, variance, Z-score, and final probability are accurate.